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A study on the reuse of remanufacturing assembly processes through the integration of multiple sources of information

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ABSTRACT

Remanufacturing is a valuable process with significant environmental benefits. However, remanufacturing companies face challenges in assembly due to small-scale production and reliance on manual labor. To address these problems and promote reuse and assembly efficiency, this paper presents a novel approach to fine-grained process retrieval and reuse, based on the process constituent elements model. The study highlights the importance of analyzing process information in four dimensions: input, auxiliary activities, value-added control activities, and output of process constituent elements. Additionally, a method for representing process multi-source information based on genetic structure is proposed. To measure similarity, the paper examines the base attributes of process constituent elements across different procedures and implements the grey relational analysis to integrate diverse information. The effectiveness is demonstrated through a set of remanufacturing assembly procedure cases.

1. Introduction

Process reuse has a significant impact on sustainability in multiple aspects. Firstly, it enhances the value of remanufacturing process knowledge, thereby boosting a company's productivity. Secondly, through the reuse of waste products and materials, process reuse extends their lifespan and minimizes the requirement for new resources. Additionally, process reuse plays a vital role in reducing carbon emissions associated with the production of new products. Consequently avoiding greenhouse gas emissions generated during the extraction and processing of fresh resources (Liao, et al., 2018). However, remanufacturing enterprises often face regulatory constraints in sourcing raw materials, primarily due to the emerging nature of the industry. Consequently, compared to traditional manufacturing enterprises, their production scale is typically smaller and relies more heavily on manual labor to complete assembly tasks (Chakraborty et al., 2021).

In the realm of remanufacturing, the assembly stage is stage is widely acknowledged as a crucial process, accounting for a significant proportion of the overall product remanufacturing workflow. Remanufacturing possesses distinctive characteristics that differentiate its assembly process from traditional manufacturing. These include small-scale production, stringent precision requirements for dimensions and fits, meticulous control of deviations and errors, advanced surface treatment and lubrication needs, as well as rigorous control and inspection standards for assembly quality. It is worth noting that manual labor plays a more prominent role in the assembly process of remanufacturing. According to research conducted by Zhu and Yao (2021), the reuse in the remanufacturing process not only enables a saving of up to 60% in energy consumption and 70% in material utilization for assembly-centric products but also achieves stable improvements in assembly quality and efficiency (Xing et al., 2022).

Therefore, to conduct thorough research on the reuse of remanufacturing assembly processes, a detailed description, exploration, analysis, and calculation of assembly process information are required. Here, the concept of "granularity" is introduced, where higher granularity indicates finer levels of detail. By decomposing the process into "product → manufacturing process plan → procedure → work step," a comprehensive understanding of the manufacturing process can be achieved, enabling the accurate capture of local detailed characteristics of the assembly process. This encompasses information on structural shape, dimensional tolerances, material properties, and heat treatment, as well as relevant approaches to manufacturing, equipment, production environment, cost-effectiveness, and traditional practices and design experience (Wu et al., 2022). The aforementioned fine-grained

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information can facilitate research on reuse in the remanufacturing assembly process.

The study investigates the retrieval and reuse of process information for remanufacturing assembly from a systemic perspective. It focuses on fine-grained process technologies and employs a systematic approach that integrates consideration, analysis, mining, and integration of information from multiple sources. Compared to existing research concerning single-method process reuse (Renu and Mocko, 2016), this study further enhances the requirements, precision, and performance of remanufacturing assembly. Through this approach, we propose a technique that calculates the similarity between the bases of the assembly procedure using the model, allowing for the effective integration of quantitative information from four process constituent elements. This method employs grey relational analysis to achieve a quantitative similarity measure of finely-grained information from various assembly procedures. The practicality and effectiveness of this technology is validated by analyzing a typical example of the remanufacturing assembly process.

The paper is organized as follows: Section 2 provides a comprehensive review of the relevant literature. Section 3 presents a research framework and representation of fine-grained remanufacturing process knowledge. In Section 4, the constituent elements of different assembly processes are quantified. Section 5 focuses on the synthesis of multi-source information. The feasibility of the theory is verified through case studies in Section 6. Finally, Section 7 concludes the paper by highlighting the limitations of the study.

2. Literature review

Current methods for process retrieval and reuse tend to be simplistic and restrictive, relying on generic process features for similarity calculations. These methods prioritize the most prominent aspects of the process, thereby limiting the efficient reuse of fine-grained process information. To overcome this limitation, it is crucial to conduct a comprehensive assessment of complete process knowledge, which necessitates the standardization of manufacturing process instructions. Furthermore, the challenge of computing similarity among disparate sources of information for various fine-grained manufacturing processes must be quantified and addressed.

2.1. Description of process knowledge

The remanufacturing process bears a strong resemblance to the manufacturing process, as both fall under the category of unstructured information. With the continuous expansion of remanufacturing industry, the number of text-based data inventories shows explosive growth, and more than 80% of enterprises choose to store manufacturing process knowledge in the form of text information (Choudhary et al., 2009). Maintaining process knowledge integrity and consistency is crucial for enhancing a firm's manufacturing intelligence. To achieve this goal, careful compilation method selection, detail level adjustments, and information content refinement are essential. This approach reduces process comprehension discrepancies among technicians within the company and ensures accurate process representation in the global distribution system. He et al. (2020) proposed a unified framework for ontology-based management of remanufacturing process knowledge, facilitating the rapid generation of remanufacturing knowledge. Yang et al. (2018) employed the Unified Modeling Language (UML) to create a standardized metamodel for manufacturing process information. This encourages compatibility and easier sharing of information between different software interfaces. Qiao et al. (2011) utilized the Process Specification Language (PSL) to define and express various aspects of the manufacturing process. It was effective for complex processing procedures. Browna et al. (1996) utilized structural grammar and its formal semantics to describe the set of objects of the manufacturing process, forming a formal manufacturing information resource for designers. Bauckhage et al. (2000) constructed a syntactic analysis approach to model the mechanical assembly category, so as to derive regular syntactic structures to make the representation of such program sets standardized and compact.

2.2. Reuse of process knowledge

Process reuse primarily involves identifying similar process knowledge by leveraging feature similarity to assist enterprises in saving time during process design and development. Wang et al. (2018) proposed an ontology for managing manufacturing process info in hot bar rolling. They demonstrated that this approach enables efficient retrieval of reusable information in design space exploration. Chen and Jia (2020) presented multimedia sources to help designers reuse assembly processes. They structure information on component topology and semantic concepts. Gong et al. (2022) proposed a method that combines case-based reasoning (CBR) and ontology to improve assembly process description by facilitating the reuse of process knowledge and assembly experience. He et al. (2020) used an ontology approach to efficiently generate remanufactured processes by reusing successful historical examples via CBR. This was demonstrated in crankshaft remanufacturing instances. Reyes et al. (2015) combined constraint satisfaction and CBR to enhance case representation and reuse. They demonstrated its efficacy in a case study involving an industrial mixing device. Li et al. (2003) developed a method to connect machining features with process data using attribute adjacency graphs, enhancing the retrieval and reuse of manufacturing information.

2.3. Similarity analysis

The core of process reuse involves analysis and calculation of similarity. Ke et al. (2020) accurately determined the similarity between remanufacturing designs with distinct features by utilizing the k-Nearest Neighbor algorithm. Fan et al. (2018) proposed a sequential method based on the edit distance metric to measure the similarity between two different processes, facilitating the transformation of sheet metal process knowledge. Renu and Mocko (2016) investigated similarity in assembly process instructions using word overlap, Jaccard score, TF-IDF, and latent semantic analysis. Wu et al. (2022) constructed a model for calculating similarity in Chinese process instructions, incorporating multiple feature combinations related to sentence dependency, semantics, structure and keyword distance. Hanifi et al. (2022) enhanced the analysis phase of inventive design by integrating Doc2vec and cosine similarity methods, using scientific data retrieval and a case study on designing a lattice structure to validate the approach. Lastra-Díaz et al. (2019) compared ontology-based semantic similarity measures and distributional measures based on word embedding models. They found that combining both approaches yielded the most successful outcomes. Han et al. (2018) constrained assembly semantically from multiple sources and introduced the weighted bipartite graph to match assembly semantics and retrieve the assembly model. Gao et al. (2015) developed a novel technique that utilizes edge-counting and information content theory to measure semantic similarity between words, surpassing human ratings and prior works. Cardone et al. (2006) found that a feature vector is crucial for evaluating the shape similarity of a part. The feature vector includes the prismatic machining direction, feature type, feature volume, feature size tolerance, and feature group base.

2.4. Summary of main issues

After conducting a literature review, the current research primarily faces three major issues:

(1) The existing techniques for describing process knowledge mainly focus on information storage and structuring in computers. The selection of an appropriate description methodology necessitates

considerations of the application context, task requirements, and data availability. While these methods provide detailed descriptions of remanufacturing assembly processes, their implementation is labor-intensive and does not facilitate intuitive content presentation.

(2) Process reuse has extensive applications in the manufacturing industry and other fields. However, adjustments are required to cater to different applications and needs, consequently increasing the complexity of reuse. The diverse standards and regulations across different objects and systems further complicate the process of reuse, demanding comprehensive considerations to overcome these limitations.

(3) The research deeply explores various similarity calculation methods applied in the field of manufacturing technologies, encompassing information text, process content, shape, product topology, and features. However, in fine-grained processes, the similarity among local features necessitates further in-depth investigation. Therefore, to construct a more comprehensive and accurate similarity calculation system, it is imperative not only to integrate multiple calculation methods but also to rigorously consider the interrelations among local features.

3. Research framework and representation of process knowledge

3.1. Research framework

In this study, we investigate the issues related to the retrieval and reuse of fine-grained remanufacturing processes by focusing on several remanufacturing assembly procedures information and establishes a specific research step. Fig. 1 depicts the framework route for our research. Our investigation begins by focusing on the assembly level process and subsequently analyzing the geometric and assembly characteristics at the granularity feature level. We propose a novel approach for multi-source information fusion utilizing the process constituent elements model. It is to qualitatively analyze, mine and research remanufacturing assembly process information from a systems management perspective.

Firstly, at the level of geometric solid lines, we can introduce process constituent elements to further dissect procedural knowledge and extract critical information. This approach involves constructing a specific base structure for processing information, considering the perspectives of points, lines, and surfaces. The objective is to develop a procedure representation model that is reusable, standardized, simplified, convenient, and "biotype". This model facilitates a systematic description of assembly process information, ensuring the organic integration procedure information from multi-sources. Next, we can achieve quantitative similarity calculation of key bases within the process constituent elements using genetic analysis. Additionally, the grey relational analysis effectively combines the quantitative information from these key bases. Finally, the feasibility of the proposed method for retrieving and reusing the information between different procedures with granularity is verified through a specific procedure example.

3.2. Representation of process knowledge

A procedure is a basic unit of the remanufacturing assembly process. Recently, Wu et al. (2022) recently proposed that any production

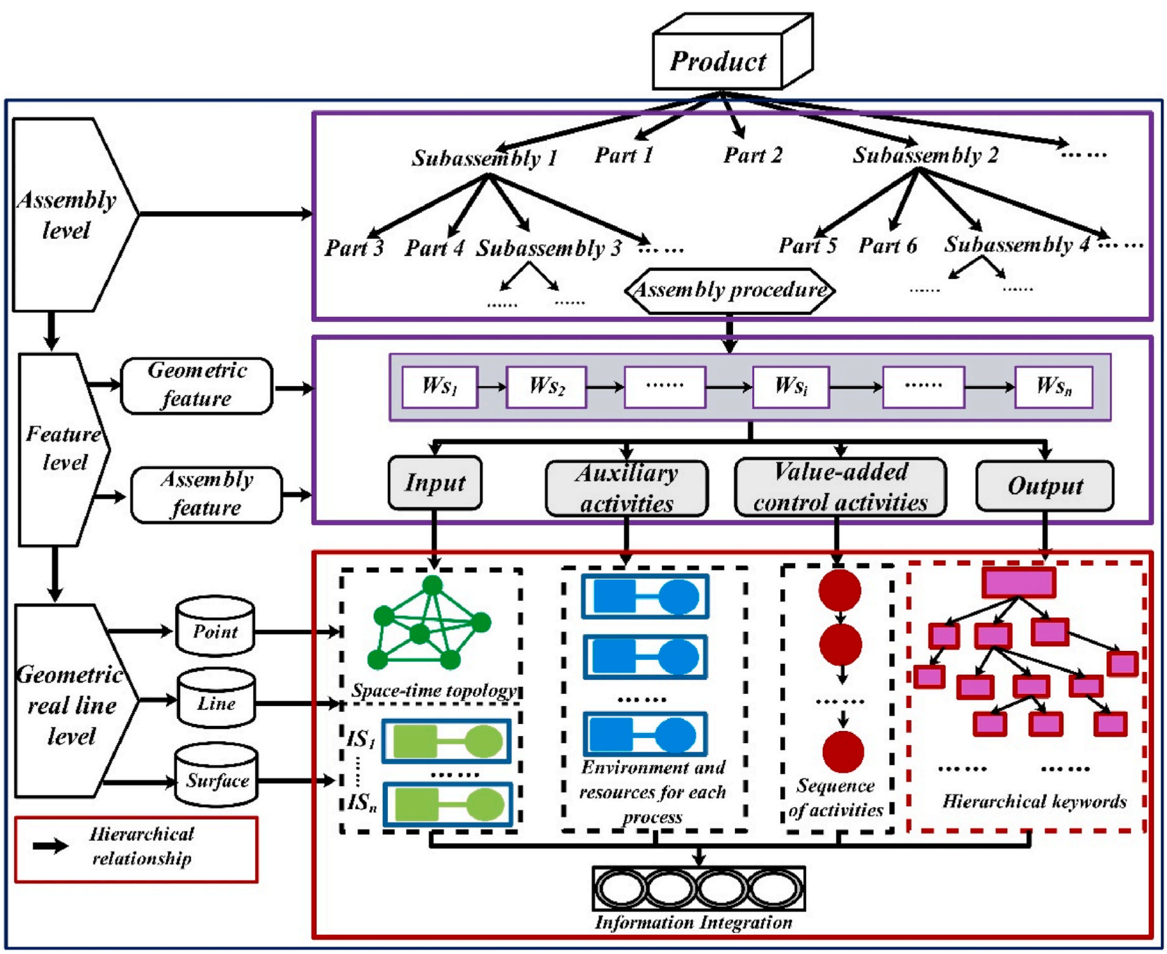


Fig. 1. Research the technical route of process.

process activity can be abstracted into a process system comprising four different categories of process constituent elements: input, process processing value-added and (quality) control activities, auxiliary activities (environment, resources), and output.

During the procedure activities, the contents expressed by the four types of process constituent elements can be summarized as follows: Input (I) specifies not only the predetermined resultant state and assembly object of the upstream process output. It also encompasses the necessary requirements for the process objectives through procedure activities. Value-added processing and (quality) control activities (VC): These activities constitute restricted procedures implemented within the procedure activities. They aim to create direct value for the product and ensure the smooth execution of activities. Auxiliary activities (AA): These activities provide essential resources such as equipment and tools, as well as the workshop environment, including temperature and humidity, necessary for the process activity. They establish a systematic production activity. Output (O) is the expected result achieved after the completion of all operation activities. Ultimately, it leads to a purposeful and conforming conclusion.

To facilitate efficient and effective process reuse, this study proposes using genetic structure to represent the inherent relationships between process constituent elements. This is because the internal fine-grained information of process constituent elements shares the following common characteristics with genetic structure:

(1) **Structural.** The constituent elements of the procedure, each with their own unique dimensional characteristics, necessitate distinct feature demands during assembly, akin to the diverse gene segments within a genetic structure that signify varying biological representations. When combined, these elements form a holistic and integrative process unit, akin to an organic entity.
(2) **Inheritance.** The essential attributes described in the procedure can be replicated, transcribed and extracted to disseminate this information to a work step or specific process constituent element.
(3) **Similarity.** The effective communication of assembly procedure information can be achieved by correlating the carrier information fractions among the process constituent elements. Once the similarity reaches a particular threshold value, it can be inferred that different assembly procedures are reusable.
(4) **Convertibility.** Process genes, analogous to biological genes, encapsulate vital information from input requisites to output outcomes, thereby forming a cohesive unit. The development of a standardized gene model can establish a uniform method for expressing process instructions.

To enhance comprehension of the whole text, Appendix A presents significant notations employed in this study.

The procedure consists of a series of uninterrupted work steps. In each dimension of the work step information, process constituent elements are sequentially extracted, and their essential features and quantitative relationships are examined. The essential information of the four distinct process constituent elements is structured in the form of gene structure sequences. Based on the concept of biological genes, this generates the Assembly Procedure Case Gene Model (APCGM) (see Equation (1)).

$$APCGM = \bigcup_{i=1}^{n} Ws_i = \bigcup_{i=1}^{n} \bigcup_{j=1}^{m} \left(I_{ij}, VC_{ij}, AA_{ij}, O_{ij}, \right) \tag{1}$$

where, $i = 1, 2, \cdots, n, j = 1, 2, \cdots, m.$

It is hypothesized that a comprehensive account of each specific manufacturing procedure can be achieved by arranging the base pairs of process constituent elements. These base-pair sequences consist of the input sequence (IS), value control sequence (VCS), auxiliary activity sequence (ACS) and output sequence (OS). The gene model sequence, as depicted in Fig. 2, effectively conveys all the critical features contained in these sequences.

$$Ws = \ < IS, VCS, AS, OS >$$

The second-level genetic structure of process constituent elements in each dimension strictly adheres to the meticulous requirements and designs defined by the respective elements (Wu et al., 2022). As mentioned in the literature (Zhao et al., 2013), the extraction, refinement, and representation of their core information as paired data can be likened to the double helix DNA structure found in genetic organisms, as illustrated in Fig. 3.

Gene bases are the basic components of the process constituent elements genes that make up each process. They can be categorized into two types: characteristic DNA chains (Chain I) and influencing DNA chains (Chain II) (see Fig. 4). Chain I comprises input attribute bases, process activity bases, auxiliary activity bases and expected result bases. Chain II consists of input management bases, process weighting bases, process influencing bases and expected outcome evaluation bases. The process constituent elements characteristic bases and process constituent elements impact bases form four base pairs in strict accordance with the principle of complementarity of base pairs.

1) Input attribute bases and input management base pairs (IB—IA): Based on the above definition of input process constituent elements, it can be seen that the base pair can include the assembly attribute rule factor specified in process operation and the space-time semantic topological structure factor of the assembly process. Fig. 4 clearly shows the key information contained in the base pair and the degree of influence, which can be expressed as Equation (2).

$$\begin{cases} \{IB - IA\} = \left(\dfrac{AA_{obj}}{\theta_{AA1}} \oplus \dfrac{AA_{req}}{\theta_{AA2}} \right) \otimes \left(\dfrac{SP_{lev}}{\theta_{sp1}} \oplus \dfrac{SP_{pos}}{\theta_{sp2}} \right) \otimes \left(\dfrac{Ti_{seq}}{\theta_{Ti1}} \oplus \dfrac{Ti_{con}}{\theta_{Ti2}} \oplus \dfrac{Ti_{adj}}{\theta_{Ti3}} \right); \\ \theta_{AA1} + \theta_{AA2} = \theta_{sp1} + \theta_{sp2} = \theta_{Ti1} + \theta_{Ti2} + \theta_{Ti3} = 1; \\ \theta_{AA1}, \theta_{AA2}, \theta_{sp1}, \theta_{sp2}, \theta_{Ti1}, \theta_{Ti1}, \theta_{Ti1} \in [0, 1] \end{cases} \tag{2}$$

where, The base pairs under the assembly attribute rule factor include AA_{obj} and AA_{req}.The space-time semantic topology factors include two parts: spatial rules and temporal rules, in which spatial rules specifically include SP_{lev} and SP_{pos}. In time rules, Ti_{seq}, Ti_{con} and Ti_{adj} are the prominent typical key information.

2) Process activity base and process weight base pairs (VB—PB): In the procedure activity, the base pair comprises an ordered set of work step value-added activity attributes and control activity attributes (See Equation (3)).

$$\{VB - PB\} = \{WS_{Vh}\} \otimes \{WS_{cv}\} \tag{3}$$

In the same procedure, each work process that can be performed independently is considered one work step. The work step value-added activities are extracted and classified using unstructured process instructions in textual format. The extracted verb information can portray the specific behavior of its value-added activities, while the base pair content of its unstructured style value-added activities is expressed as Equation (4).

$$\begin{cases} \{WS_{vh\_attr}\} = \dfrac{ws_{vh1}\_word}{\theta_{h1}} \oplus \dfrac{ws_{vh2}\_word}{\theta_{h2}} \oplus \cdots \oplus \dfrac{ws_{vhn_l}\_word}{\theta_{hr}} \cdots \\ \displaystyle\sum_{r=1}^{n_l} \theta_{hr} = 1, \theta_{hr} \in [0, 1] \end{cases} \tag{4}$$

where $r = 1, 2, \cdots n_l$, n_l represents the number of verbs in all steps of the procedure.

Control activity base pairs are primarily implemented to ensure the smooth implementation of each work-step process, aiming to minimize

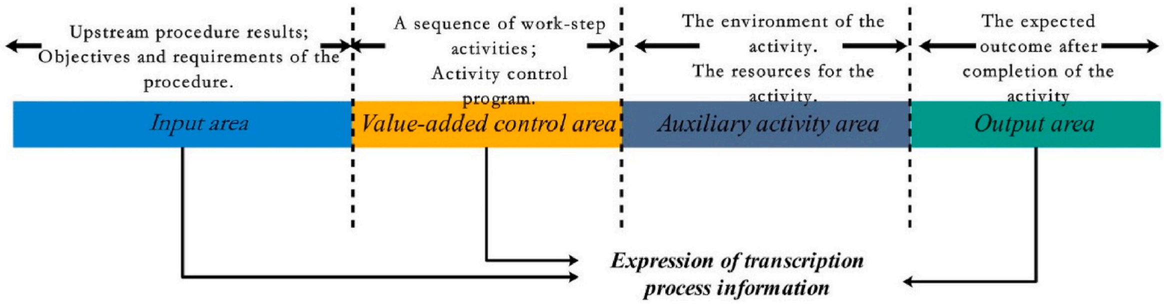


Fig. 2. Gene model of the operation sequence.

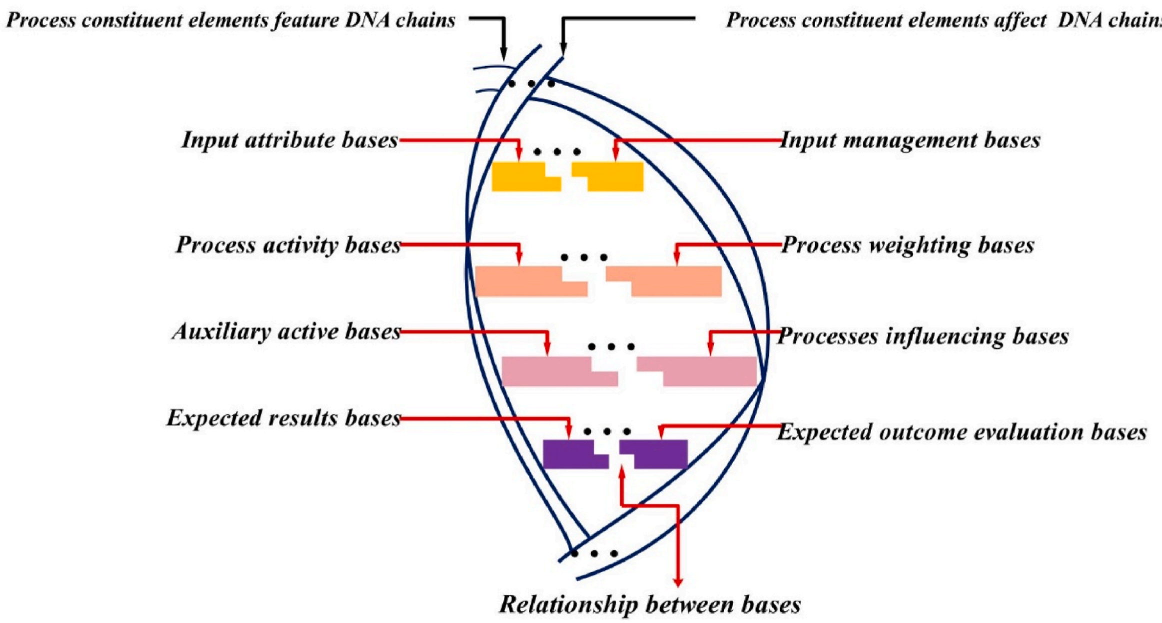


Fig. 3. Gene DNA structure of the process constituent elements of the procedure.

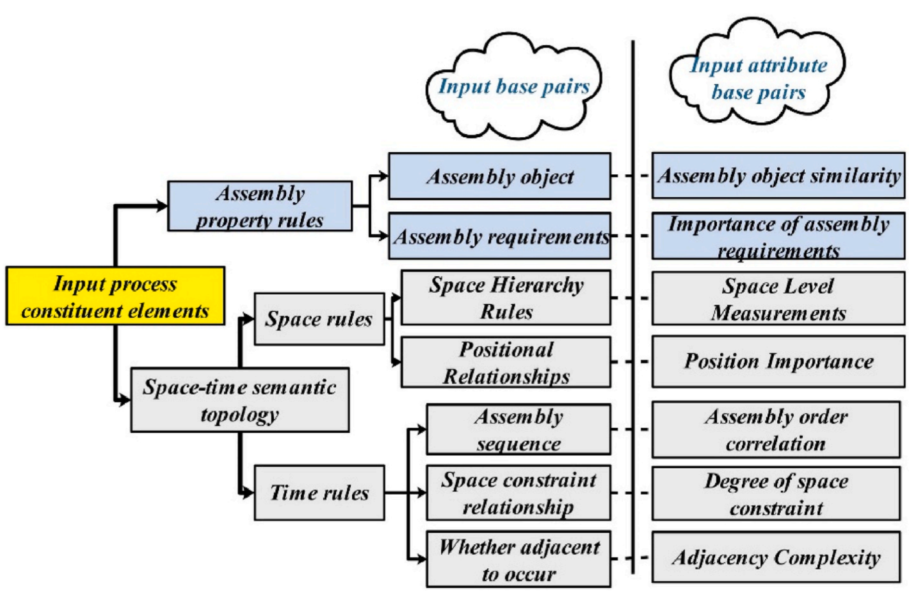


Fig. 4. Key information of input base pairs.

potential failure modes. These pairs emphasize two types of control behavior: preventive factors and identification factors. Preventive factors aim to eliminate failure causes or failure modes and reduce their occurrence rate. Identification factors aim to identify potential process failures and their underlying causes, guiding the development and enhancement of corrective measures. These factors are expressed as follows Equation (5).

$$\begin{cases} \left\{WS_{cv\_attr}\right\} = \dfrac{WS_{vhp}\_trpe}{\delta^h_{cv1}} \otimes \dfrac{WS_{cvi}\_type}{\delta^h_{cv2}} \\ \delta^h_{cv1} + \delta^h_{cv2} = 1 \quad \delta^h_{cv1}, \delta^h_{cv2} \in [0, 1] \end{cases} \tag{5}$$

3) Auxiliary activity bases and processes affect base pairs ($AcB - PaB$): The main consideration is the required resource factors and environmental factors in the process activities. To ensure product compliance with process requirements, it's crucial to assess and oversee the work environment, equipment, and tools needed for carrying out the activities. The base pair can be expressed as Equation (6).

$$\{AcB - PaB\} = \{Re_{s1}, Re_{s2}, \cdots, Re_{sw}\} \otimes \{En_{s1}, En_{s2}, \cdots, En_{sg}\} \tag{6}$$

4) Expected outcome bases and expected outcome evaluation base pairs ($ErB - EaB$): Each work-step activity has a varying degree of influence on the final outcome of the completed process; this relationship can be represented as a base pair (see Equation (7)).

$$\{\mathrm{ErB} - \mathrm{EaB}\} = \frac{\mathrm{ER}_{s1}}{\theta_{\mathrm{ERs}_1}} \oplus \frac{\mathrm{ER}_{s2}}{\theta_{\mathrm{ERs2}}} \oplus \cdots \oplus \frac{\mathrm{ER}_{si}}{\theta_{\mathrm{ERsi}}} \oplus \cdots \tag{7}$$

4. Quantitative measures of gene bases of process constituent elements

Through qualitative analysis of genes associated with diverse fine-grained assembly procedures, the unified gene structure representation method facilitates the incorporation of knowledge pertaining to each process constituent element. The genes governing each process constituent element are made up of diverse base pairs, perceived as fundamental fragments of process information present in the constituent elements, influencing the procedure to varying degrees. To assess the similarity between genes of different process constituent elements, it is essential to model and solve each base. Therefore, this section proposes the development of explicit quantitative metrics to evaluate the DNA structures of distinct procedural genes.

4.1. Input process constituent elements

The input process constitute elements is fundamental that are essential to the assembly process. The analysis of four specific local similarity factors is necessary: assembly object, assembly requirements, assembly space rules, and assembly timing rules. The accurate identification and selection of components, materials, and parts require a comprehensive approach. Additionally, the required regulations for the assembly process, the spatial arrangement of objects, and the sequencing of assembly components must also be considered.

(1) Assembly object.

The assembly object is a crucial component of any process unit, but the entity types of its assembly object vary across different procedures. When mining and reusing fine-grained procedure information, determining the object similarity requires more than a simple Boolean matching operation based on text structure names. Instead, it relies on examining the properties and functions of assembly objects, As shown in Fig. 5 (Rodriguez and Egenhofer, 2003). Building upon the reference

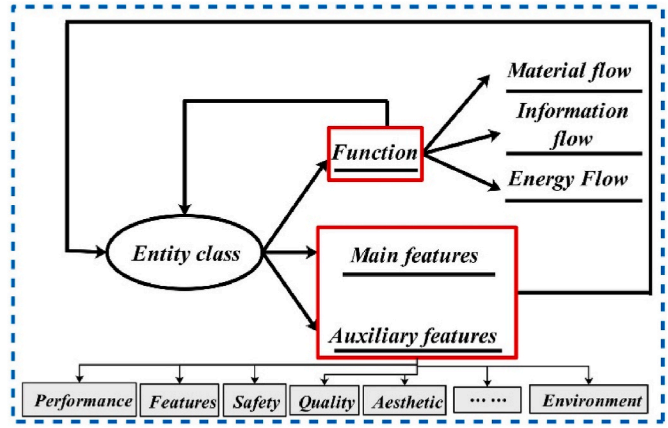


Fig. 5. Entity features of assembly objects in the input process constituent elements.

(Deng, 2002), a principle for measuring the similarity metrics between different entity objects based on two factors has been proposed: basic functions and demand characteristics of assembly objects. The functions can be categorized into three main categories: material flow, information flow, and energy flow. Conversely, demand characteristics encompass 12 key features, such as performance, characteristics, safety, quality, and aesthetics. The importance of these features varies among different assembly objects. Assuming two assembly objects, AA^1_{obj} and AA^2_{obj}, the similarity between them can be determined using Equation (8).

$$\mathrm{Sim}\left(\mathrm{A}^1_{\mathrm{obj}}, \mathrm{A}^2_{\mathrm{obj}}\right) = \frac{1}{2}\frac{2 \times \mathrm{Fun}_{\mathrm{Num}}}{\mathrm{A}^{1\mathrm{Fun}}_{\mathrm{obj}} + \mathrm{A}^{2\mathrm{Fun}}_{\mathrm{obj}}} + \frac{1}{2}\left(\frac{2}{3}\left(\frac{2 \times \mathrm{Num}^{\mathrm{P}}}{\mathrm{A}^{1\mathrm{p}}_{\mathrm{obj}} + \mathrm{A}^{2\mathrm{p}}_{\mathrm{obj}}}\right) + \frac{1}{3}\left(\frac{2 \times \mathrm{Num}^{\mathrm{Q}}}{\mathrm{A}^{1\mathrm{Q}}_{\mathrm{obj}} + \mathrm{A}^{2\mathrm{Q}}_{\mathrm{obj}}}\right)\right) \tag{8}$$

where, A^{1Fun}_{obj} and A^{2Fun}_{obj} are the number of functional features listed from three perspectives, Num^P and Num^Q are the common number of primary and secondary features of the two assemblies among the 12 requirement features, A^{1p}_{obj} and A^{1Q}_{obj} are the main requirement features of the two assemblies, A^{2p}_{obj} and A^{2Q}_{obj} are the auxiliary features of the two different assemblies.

(2) Assembly requirements.

When considering assembly requirements, it is essential to select indicators that not only directly reflect the assembly conditions but also employ reliable surrogate indicators that signify assembly performance. These indicators may comprise surface roughness, assembly feature types, accuracy levels, size specifications, assembly parameters, and other additional technical requirements. These requirements should be based on the specific circumstances of the assembly under consideration. It is worth noting technical specifications can be illustrated in addition to assembly drawings, as demonstrated in Fig. 6.

To determine the similarity between two distinct assembly requirements, it is essential to extract the typical features from their semantic content. The calculation of similarity between the typical features across various attributes should follow a standardized measure. This measure needs to be analyzed and applied to each type of data throughout the remanufacturing process.

Type1: similarity of assembly shape structure. The maximum length of parts in two different assembly processes is compared and calculated using the similarity measure provided in Equation (9)

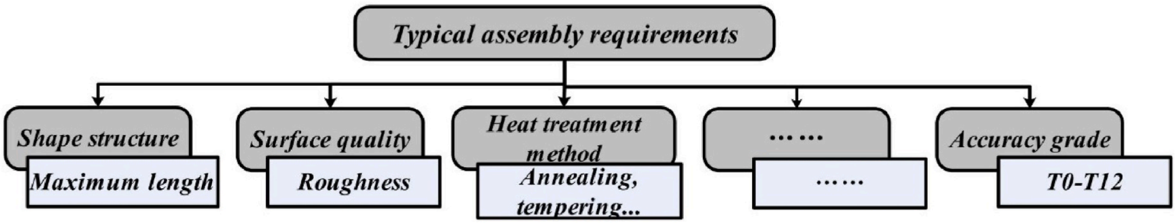


Fig. 6. Typical indicators in the assembly requirements of the input process constituent elements.

$$\mathrm{Sl} = 1 - \frac{|\mathrm{L_i} - \mathrm{L_j}|}{\max(\mathrm{L}) - \min(\mathrm{L})} \tag{9}$$

Type2: Roughness similarity calculation. In the process of assembly process planning, roughness is closely related to the assembly operation parameters. When comparing two assembly features with similar procedures, a smoother surface is achieved with smaller surface roughness. The standard series of roughness for assembled parts can be approximated as a proportional series. Therefore, a fuzzy subordination function can be formulated as shown in Equation (10).

$$u_{Ra}(x_i) = \begin{cases} 0 & Ra < Ra_L(x_i) \\ 1 - \dfrac{(lnRa_H(x_i) - lnRa)}{2(lnRa_H(x_i) - lnRa_L(x_i))} & Ra_L(x_i) \le Ra \le Ra_H(x_i) \\ 1 & Ra > Ra_L(x_i) \end{cases}. \tag{10}$$

Type3: Similarity of heat treatment methods of assembled parts. The similarity between parts P and Q of two different procedures, with different heat treatment methods, the operation relation between them can be regarded as Boolean operation, and the similarity is expressed by Equation (11).

$$S_{ht} = \begin{cases} 0 & P = Q \\ 1 & P \ne Q \end{cases} \tag{11}$$

Type4: Precision level similarity. If accuracy levels are expressed as interval numbers, it is necessary to represent the range of the interval using spatial points. The similarity between these points should be evaluated by measuring the distance between them, whereby larger distances indicate lower degrees of similarity. If $p_{ij} = [p_{ij}^l, p_{ij}^u]$ and $p_{0j} = [p_{0j}^l, p_{0j}^u]$, the distance is transformed into a similarity value by applying the Euclidean distance method, as shown in Equation (12).

$$S_{ac} = 1 - \frac{\sqrt{\left(p_{ij}^l - p_{0j}^l\right)^2 + \left(p_{ij}^u - p_{0j}^u\right)^2}}{max\sqrt{\left(p_{ij}^l - p_{0j}^l\right)^2 + \left(p_{ij}^u - p_{0j}^u\right)^2}}. \tag{12}$$

(3) Space rules

The requirements for input process constituent elements of the assembly procedure encompass the attributes of the assembly object properties, assembly requirements and specific attributes, numerical information, as well as the geometric and structural relationships specified during the assembly process. To effectively capture and describe these relationships, we propose spatial semantic topological rules. These rules serve to meet the criteria for capturing and representing the input elements of the assembly process, ensuring the accuracy and comprehensiveness of the description. The spatial rules define the spatial location and constraint relationships among the assembled parts. Prior to assembly, it is essential to fulfill the geometric positioning relationship between parts in space. Assembly constraints, which represent the fundamental unit relationships enabling local functionality during the assembly process, emphasize the interdependencies among parts. These constraints reflect the product's function and are determined by specific assembly characteristics. To describe the structure of parts contributing to specific functions in the assembly process, the concept of space structure semantics is introduced. This facilitates the expression of abstract semantics suitable for conveying information on product assembly space design. The structure semantics of space can be categorized into three types: connection semantics, motion semantics, and positioning support semantics, as depicted in Fig. 7.

Considering spatial constraints in detail is a time-consuming and error-prone process. However, three common design features can be identified and classified as baseline, positional, and contour features. These features can be utilized to construct the planar feature outlines of complex structures, which can then be manipulated using techniques such as stretch, sweep, rotation, and deflation to generate 3D features. Therefore, this paper focuses on spatial rules for two-dimensional constraint relationships, as detailed in Appendix B.

To develop a suitable model for calculating the similarity of assembly procedure space constraints, the combination of assembly semantics and constraint relationships should be considered as similar elements, while similar entities in fit relationships should be viewed as similar objects.

Let the assembly semantics be the first similar element with the scale factor set to e_{1j} , and the constraint relation be the second similar element with the scale factor set to e_{2j} , as for the constraint relation, the degree of similarity depends mainly on the difference between the number of features k and l, as shown in Equation (13).

$$\begin{cases} q(u_1) = e_{1j}(\text{Assembly semantic}) \\ q(u_2) = \dfrac{m}{k + l - m}(\text{Constraint semantic}) \end{cases} \tag{13}$$

where, the assembly input process constituent elements a_i contains k typical features, b_i contains l typical features, and the two input process constituent elements a_i and b_i contain m common similar features.

A similarity algorithm is developed for the fit relationship, assuming equal weights for each comparison element. The algorithm is based on the semantic expression of the assembly and the definition of the assembly constraint relationship. The calculation for this algorithm is given by Equation (14).

$$q(u_i) = q(u_1)q(u_2) = \frac{m}{k + l - m}\sum_{i=1}^{m}\frac{1}{m}q(u_i). \tag{14}$$

(4) Time rules

The time rules determine the matching features of the assembly process, and the assembly sequence of the units is determined by the assembly constraints. It also involves judging the assembly base parts and the knowledge of whether the assembly actions occur adjacently. The product flow in process operations was analyzed, and the consideration of assembly process events as temporal objects was proposed. The temporal state of assembly was represented using time points and time periods. Temporal topological relations are introduced to describe the temporal semantic knowledge of assembly process events in assembly planning. This temporal semantic knowledge was presented using the topological representation of ontology (Chen et al., 2020). By establishing an explicit association between these elements, a standardized and unified expression is achieved. Fig. 8 illustrates the

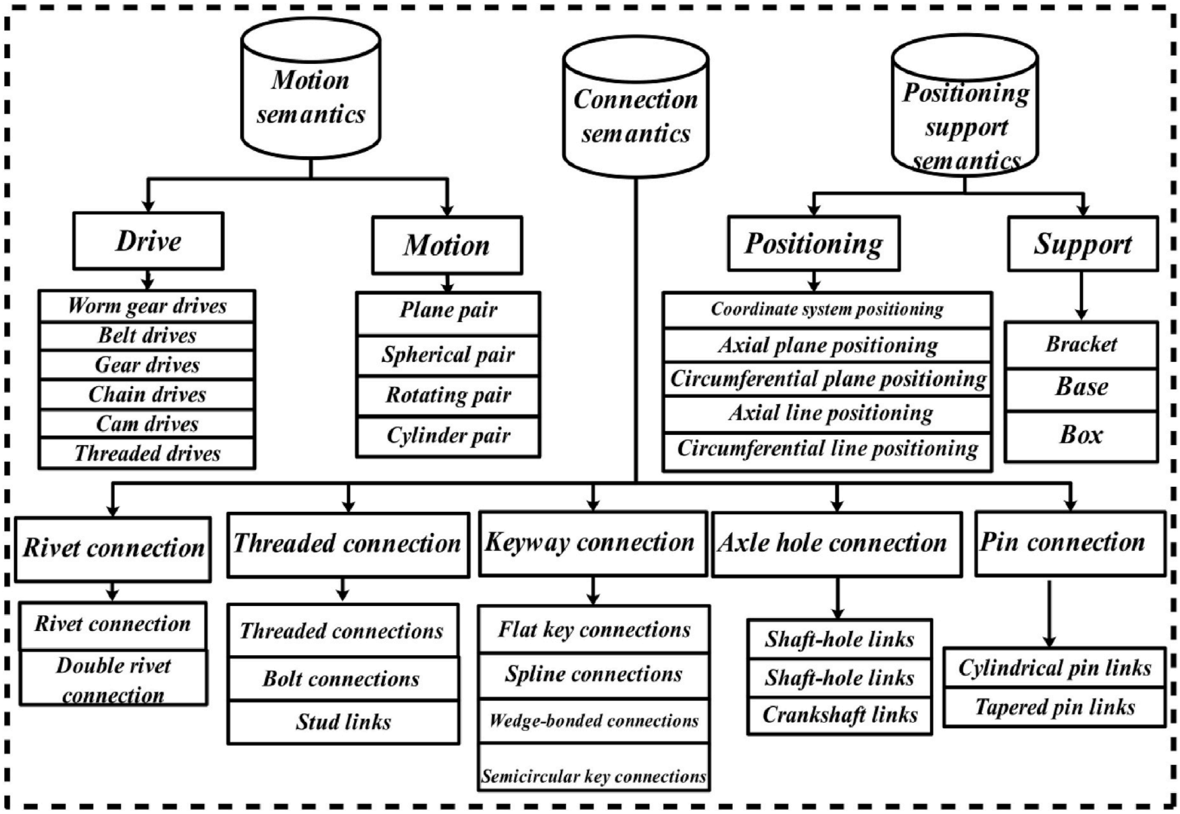


Fig. 7. Extraction of assembly semantics.

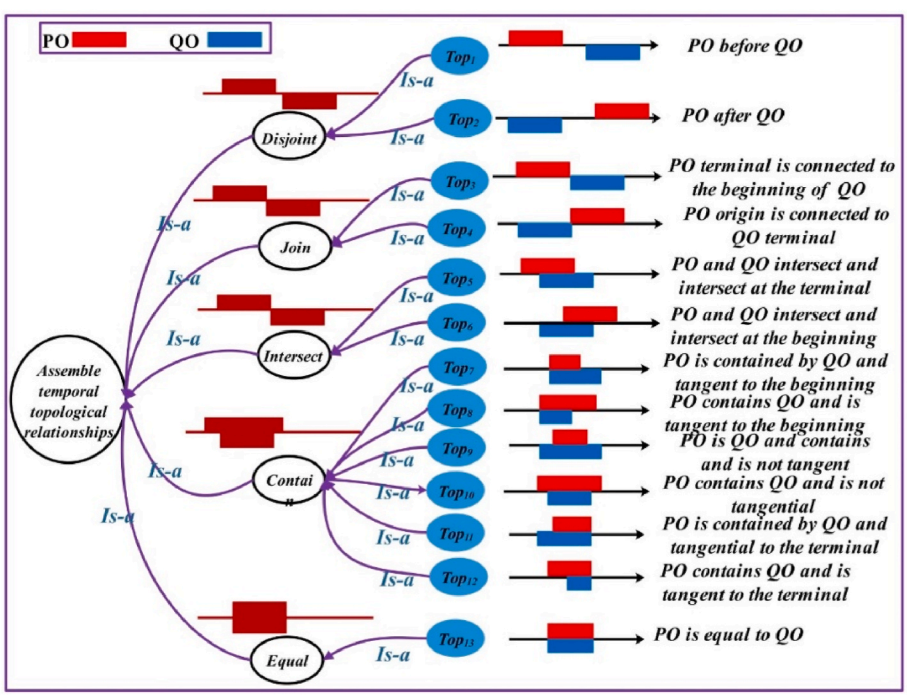


Fig. 8. Temporal topology of the products in the input process constituent elements.

ontology relations constructed in this approach. Additionally, 13 temporal topological relations are derived based on the five basic geometric topological relations.

The ontology can represent unstructured knowledge and achieve a unified measure of similarity between different process operations within its framework. Following the measure relationship proposed by Li et al. (2003), we combine the shortest path length with the depth of ontology information to measure the similarity relationship between different assembly procedure time topologies, as presented in Equation (15)

$$Sim_i(P,Q) = e^{-\alpha \min|path_i(P,Q)|} \bullet \frac{e^{\beta h} - e^{-\beta h}}{e^{\beta h} + e^{-\beta h}} \quad (15)$$

where, α and β are the contribution parameters of the shortest path length and depth under the ontology path, respectively. The experiments show that the optimal parameters are set to $\alpha = 0.2$; $\beta = 0.6$.

4.2. Auxiliary activities

The resources in auxiliary activities serve as material guarantees for the smooth completion of assembly processes. These resources include connected objects, auxiliary tools and equipment. If standardized equipment and tools are used for each process activity, the tools and equipment involved in each process resource activity can form a set, and the similarity issues between different processes can be transformed into similarity issues between two sets of processes. By adopting a similarity model based on set theory, similarity description is achieved through a feature matching process. This process takes into account the linear combination of common and unique features among objects. In practice, for calculating the similarity between two sets. In practice, the study by Solé-Ribalta et al. (2014) demonstrates that calculating similarity using set theory generally provides more accurate results (see Equation (16)). If F_{a_R} and F_{b_R} denote the set of resources of two different assembly procedures, respectively,

$$Sim_{resource}(F_{a\_R}, F_{b\_R}) = 1 - log_2\left(1 + \frac{|F_{a\_R} \backslash F_{b\_R}| + |F_{b\_R} \backslash F_{a\_R}|}{|F_{a\_R} \cup F_{b\_R}|}\right) \tag{16}$$

where, considering the different classification characteristics of the two sets, $F_{a\_R} \backslash F_{b\_R}$ is called the difference degree, which is calculated as follows:

$$F_{a\_R} \backslash F_{b\_R} = F_{a\_R} - F_{b\_R} = \{c \in F_{a\_R} \wedge c \notin F_{b\_R}\}$$

Effective management and control of complex environmental changes within the workshop are necessary requirements in the remanufacturing process. Temperature is often a primary consideration in remanufacturing activities. Environmental factors involved in different remanufacturing procedures are typically defined using interval values to meet manufacturing condition requirements. For instance, "room temperature" is commonly defined as a temperature range of [16 °C, 30 °C]. The issue of similarity between interval values of different influencing factors can be addressed through integral transformation. If the interval values of a specific environmental factor in two procedures are denoted as ' $[a_1, a_2]$' and ' $[b_1, b_2]$' respectively, Equation (17) for calculating similarity is as

$$sim([a_1, a_2], [b_1, b_2]) = \frac{\int_{a_1}^{a_2}\int_{b_1}^{b_2} 1 - |y - x| \Big/ (U - T)\mathrm{d}y\mathrm{d}x}{(a_2 - a_1)(b_2 - b_1)} \tag{17}$$

where $a_1, a_2, b_1, b_2 \in [\mathrm{T,U}]$, it is obvious that Equation (17) is a segmented function after taking to absolute values and its integral result is related to the magnitude of the upper and lower limits of $[a_1, a_2]$ and $[b_1, b_2]$.

The similarity of the environmental factors between the two intervals for different procedures is obtained after the classification discussion (see equation (18)).

$$sim([a_1, a_2], [b_1, b_2]) = \begin{cases} 1 - \dfrac{(b_2 + b_1 - a_2 - a_1)}{2(U - T)} & a_2 \le b_1 \\ 1 - \dfrac{3(b_2 - b_1)(b_1 - a_1)(b_2 - a_1)}{6(a_2 - a_1)(b_2 - b_1)(U - T)} + \dfrac{(a_2 - b_1)^3 + (a_2 - b_2)^3 + (b_2 - b_1)^3}{6(a_2 - a_1)(b_2 - b_1)(U - T)} & b_1 \le a_2 \le b_2 \\ 1 - \dfrac{b_2^2 + b_1^2 + b_2 b_1}{3(a_2 - a_1)(U - T)} - \dfrac{a_2^2 + a_1^2 - (a_2 + a_1)(b_2 + b_1)}{2(a_2 - a_1)(U - T)} & b_2 < a_2 \end{cases} \tag{18}$$

where, if $a_1 > b_1$, the values of a_1, b_1 and a_2, b_2 can be interchanged according to the derivation process to obtain a similar similarity to Equation (18).

4.3. Value-adding & control activities

In the activity of assembly procedures, the operational process can be divided into multiple work steps. One of the most common structures for assembly process instructions is the "predicate + object" format, where the predicate is generally a verb and the object is the object of assembly. To tackle issues related to varying terminology and poor consistency in these instructions, the verbs utilized in the sentence structure are extracted based on the sequence of work-step instructions. These extracted verbs are then standardized according to a standardized functional ontology structure. The standardized verbs for various process activities are as

$x_1 = \{ox_1, ox_2, ox_3, \cdots, ox_{n1}\}$ and $x_2 = \{ox_1^{'}, ox_2^{'}, ox_3^{'}, \cdots, ox_{n2}^{'}\}$.

If $s(i,j)$ is the score of the first i and the first verb j sequences of the extracted sequence x_1 and x_2 of value-added work-step activities, the most classical Needleman-Wunsch algorithm (Day, 2010) is adopted to solve its similarity when dealing with the global similarity problem of the two sequences, and its maximum value is calculated utilizing Equation (19)

$$s(i,j) = max\begin{cases} 0 & (i = j = 0) \\ s(i-1,j) + \delta(i,-) & (j = 0) \\ s(i,j-1) + \delta(-,j) & (i = 0) \end{cases} \tag{19}$$

where $\delta(i,j)$ is called the score function and "$-$" indicates an empty space. When the value-added control activities are perfectly matched, it is recorded as $+1$; when there is a mismatch, it is recorded as 0, when an empty space is inserted, it is recorded as -1. The details are given by Equation (20).

$$\delta = \begin{cases} 1 & (os^{'} = os \neq -) \\ -1 & (os \neq os^{'}, os \neq -, os^{'} \neq -) \\ -1 & (os = -or\ os^{'} = -) \end{cases} \tag{20}$$

The initial conditions for the calculation are given by Equation (21).

$$s(i,j) = \begin{cases} 0 & (i = j = 0) \\ s(i-1,j) + \delta(i,-) & (j = 0) \\ s(i,j-1) + \delta(-,j) & (i = 0) \end{cases} \tag{21}$$

where, there are three recursive paths in the recursive calculation of $S(i,j)$ values, which go back along the lower right corner to the upper left corner, with three recursive positions for each locus: upper left, left and upper edge. When the backtracking path is along the diagonal, the verbs indicating the current value-added control activity can be paired with each other. If the path is up or to the left, it indicates that the verb of a value-added activity is missing at that position and needs to be inserted into the empty space.

The similarity of the value-added activities of two different procedures can be calculated by the following Equation (22)

$$Sim\ (x_1, x_2) = \frac{|x_1 \cap x_2| + G}{|x_{max}| + g}. \tag{22}$$

where x_{max} denotes the global best match of x_1 and x_2; $|x_{max}|$ denotes the length of the global best match; $|x_1 \cap x_2|$ denotes the number of verbs that can be matched; and G denotes the reward function for successive identical value-added activities to obtain successive matches in the calculation process, as presented in Equation (23).

$$G = \frac{1}{2}\sum_{i=1}^{Z}(|e_i| - 1) \tag{23}$$

where Z denotes the number of consecutive value-added predicate verb matches; e_i denotes the *i-th* consecutive matching process.

The penalty function in the case of blank space penalty is calculated as shown in Equation (25).

$$g = \frac{1}{3}\sum_{j=1}^{v}\left(|b_j| - 1\right) \tag{24}$$

where b_j is the number of the *j-th* consecutive vacancy and denotes the number of missing consecutive value-added predicate verbs.

The primary objective of control activities is to ensure quality of each remanufacturing stage. The vast majority of the sub-processes are visually, tactually, or auditorily inspected or verified employing auxiliary tools to ensure effective process control. However, assessing the level of maturity and detection opportunities in these approaches is a qualitative process, characterized by fuzziness, which can be described using fuzzy linguistic properties. In general, the levels of fuzziness can be categorized as, very low [0–19], low [20–39], medium [40–59], high [60–79], and very high [80–100]. The corresponding membership functions are shown in Fig. 9. The similarity between two fuzzy language terms can be calculated based on Equation (25)

$$Sim(e_{ij}, e_{0j}) = 1 - \frac{\int_\alpha^\beta \int_\alpha^\beta u(x)u'(y)|x-y|dxdy}{(\beta-\alpha)\int_\alpha^\beta u(x)dx \int_\alpha^\beta u'(y)dy} \tag{25}$$

where, $u(x)$ and $u'(y)$ denote two different language-valued membership functions, α and β are the lower and upper bounds of the range, respectively.

4.4. Expected results

The expected result is the conclusion of the process output, which is described semantically through concise textual sentences that indicate the expected qualitative results and comments achieved by the remanufacturing process. From a natural language perspective, studying the similarity between different procedure outputs essentially involves examining the similarity between the core semantics expressed by the short textual sentences with different output conclusions. In their work (Wu et al., 2022), the authors highlight that complete sentences consist

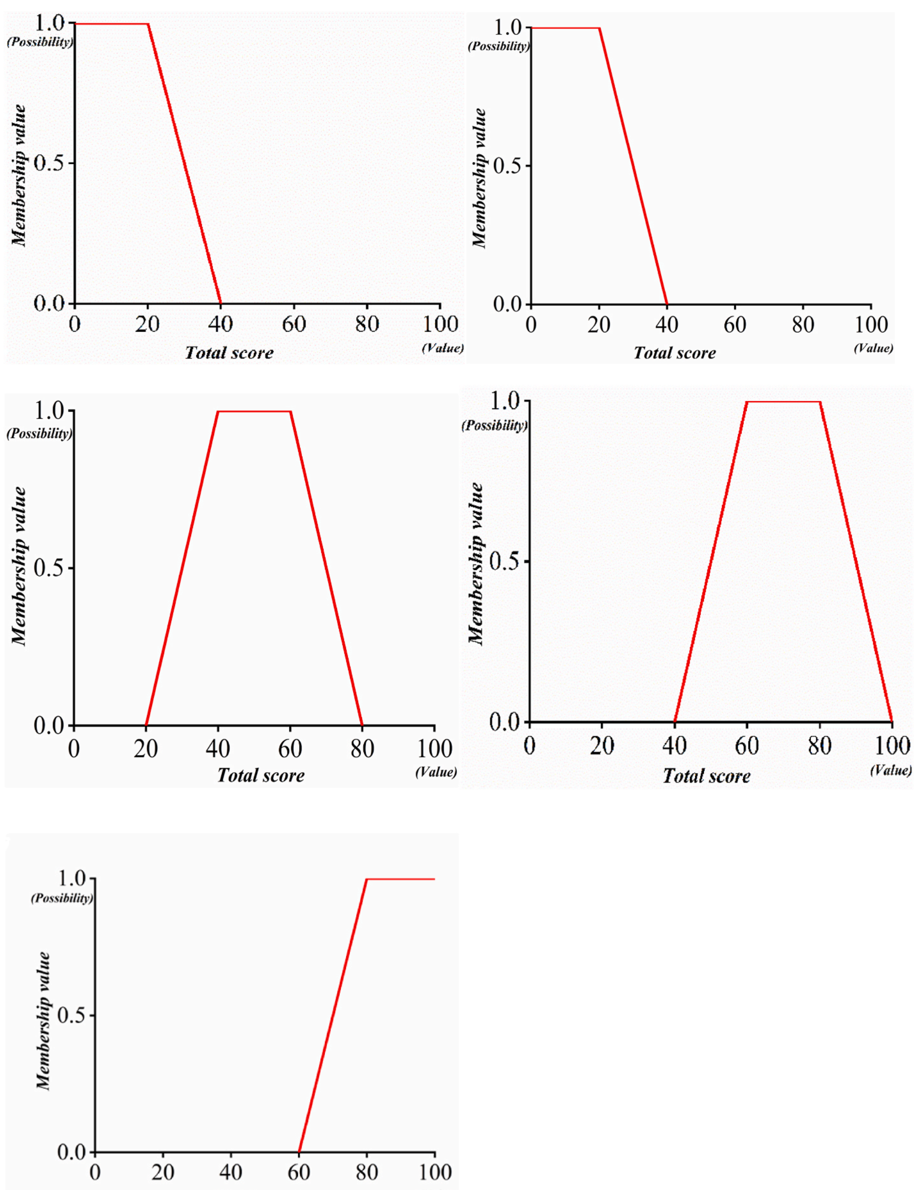


Fig. 9. Membership at different language levels.

of a central word, obligatory arguments, and optional arguments. The similarity between the expected outcomes of the outputs of two different procedures can be considered as the sum of the similarities between the key objects of two sentences with the same syntactic function (Oliva et al., 2011). This similarity is calculated as shown in Equation (26).

$$Sim(s_1,s_2)=\frac{1}{n}\sum_{i=1}^{n}sim(h_{1i},h_{2i})-\eta\times PF. \tag{26}$$

where, if sentence s_1 is composed of n phrases which are $h_{11},\cdots,h_{1n}$ and sentence s_2 is composed of n phrases which are $h_{21},\cdots,h_{2n}$,respectively. The phrases h_{1i} and h_{2i} have the same syntactic function. η is the number of syntactic roles in both sentences that appear in only one of the sentences. η reflects the fact that only one of the two sentences has the phrase and the other does not have the phrase, and a penalty factor PF is introduced to quantify this fact with additional information (Huang and Cui, 2014).

5. Synthesis of multi-source information

The process constituent elements and their gene base structures effectively and systematically represent the information content of the procedure. To compare the similarity of different assembly procedures, it is necessary to synthesize information from multiple sources regarding the DNA structure of four types of process constituent elements. Currently, experts evaluate and make decisions on each attribute information separately, determine attribute weights, and then combine them to obtain an overall similarity value. However, this method fails to consider the independence of the process constituent elements and base information. To address this issue, this study introduces the grey relation analysis for calculating comprehensive attribute similarity. The core idea is to determine the degree of relation degree between two procedures based on the similarity trend of the four process constituent elements. Consistency in the trend indicates a higher relation, while inconsistency indicates a lower relation.

To efficiently achieve the retrieval and reuse of assembly procedures at a fine-grained level, it is necessary to establish a sequence of attributes $A=\{a_1,a_2,a_3,\cdots,a_n\}$ for analyzing the characteristics of the process constituent element. Furthermore, the new procedure to be inspected should be matched with other instances of procedures, serving as the target task, thereby forming a set of tasks to be retrieved. If the target task is Obj, and the set of tasks to be retrieved is $CP=\{cp_1,cp_2,cp_3,\cdots,cp_m\}$, The matrix for calculating similarity is shown in Equation (27).

$$s=\begin{Bmatrix} Sim(Obj(a_1),cp_1(a_1)) & Sim(Obj(a_1),cp_2(a_1)) & \cdots & Sim(Obj(a_1),cp_m(a_1)) \\ Sim(Obj(a_2),cp_1(a_2)) & Sim(Obj(a_2),cp_2(a_2)) & \cdots & Sim(Obj(a_2),cp_m(a_2)) \\ \vdots & \vdots & \ddots & \vdots \\ Sim(Obj(a_n),cp_1(a_n)) & Sim(Obj(a_n),cp_2(a_n)) & \cdots & Sim(Obj(a_n),cp_m(a_n)) \end{Bmatrix} \tag{27}$$

where, $Sim(Obj(a_i),cp_j(a_i))$ is the similarity between the target task Obj and the attribute a_i of the task cp_j to be retrieved.

The specific steps of information fusion for grey relation analysis are as follows:

Step 1: The classic analytic hierarchy process (AHP) is applied to quantitatively assess the importance of independent specific base pairs within the same process constituent elements and determine their weight corresponding coefficients.

Step 2: Taking the maximum value of each row of the similarity matrix s as the reference value for the grey relation analysis, the reference series can be transformed into

$$R=\{r_0(1)=max(a_1),r_0(2)=max(a_2),\cdots,r_0(n)=max(a_n)\}$$

Step 3: The comparison sequence corresponding to the set of tasks to be retrieved can be expressed

$$s_j=\{s_j(1),s_j(2),\cdots,s_j(i),\cdots,s_j(n)\}$$

Step 4: Calculate the number of grey relation coefficients between $r_0(i)$ and $s_j(i)$. As presented in Equation (28).

$$\zeta_{0j}=\frac{\min_j\min_i|r_0(i)-s_j(i)|+\rho\max_j\max_i|r_0(i)-s_j(i)|}{|r_0(i)-s_j(i)|+\rho\max_j\max_i|r_0(i)-s_j(i)|} \tag{28}$$

where, $\rho\in[0,1]$ is the resolution coefficient, $\rho=0.5$, and $\min_j\min_i|r_0(i)-s_j(i)|$ and $\max_j\max_i|r_0(i)-s_j(i)|$ are the minimum and maximum differences of the two levels, respectively.

Step 5: Calculate the degree of relation. The larger the number of attribute relation coefficients resulting from the synthesis of different process constituent elements, the closer the relation between tasks and the greater the similarity (See Equation (29)).

$$C(s_j)=\frac{1}{n}\sum_{i=1}^{n}\zeta_{0j}(i) \tag{29}$$

6. Case application

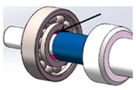
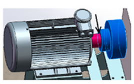
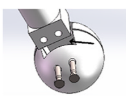
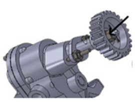
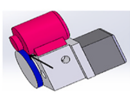
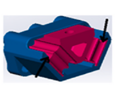
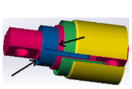
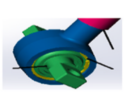
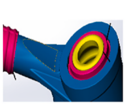
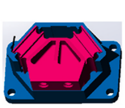
Remanufacturing assembly is the process of assembling products according to process design requirements to meet technical specifications, precision, and performance. This study takes the example of reassembly activities in a specific company's remanufacturing practice, where the proposed method is applied to the assembly process of 14 remanufactured parts' assembly, intending to validate its effectiveness. Table 1 presents statistical data related to the assembly information of the component assembly process.

Take (AF1, AF2) as an example, in the input process constituent elements, based on Fig. 5, the assembly object "rolling bearing" and "coupling" functions and features (major and minor features) are considered respectively, and the similarity between them can be obtained in accordance with Equation (8). The value of similarity between them is 0.6875. In typical assembly requirements, the length is used to measure the shape of the assembled parts, a fuzzy membership function is applied to measure roughness, Boolean operation is employed to solve the heat treatment method, and the assembly accuracy level is measured using intervals. By combining these measurements with Equations (9)–(12), similarity values of 0.7500, 1.0000, 1.0000, and 0.8837 are obtained.

The spatial rules between rolling bearings and shafts, as well as couplings and shafts, in the assembly semantics follow a shaft-hole connection type. In the constraint relationship, the mounting holes of bearings and couplings align collinearly with the axis, while the two assembly objects align collinearly with the bus bar of the shaft. Additionally, the two parts overlap with the axial end face respectively. However, when connecting the coupling and the shaft, it is necessary to ensure that the side of the flat key and the coupling aperture are coplanar. Therefore, by Equations (13) and (14), we obtain the constraint relation on $k=2$, $l=3$, $m=2$, $q(u_2)=2/3$, resulting in a similarity value of the spatial rule as 0.8333.

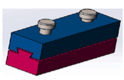
On the time rules, as in Fig. 8, AF1 belongs to Top5, AF2 belongs to Top6, $path=3$ and $h=3$, with a similarity value of 0.9468 as per Equation (15). In this research paper, the assembly procedure of the part, and the impact of temperature on it are considered in the auxiliary activities. The similarity of the environmental factor can be obtained from Equation (18) as 0.2596, based on the requirements of temperature interval changes. The similarity between resources is calculated by using

Table 1
Statistical analysis of critical information during the assembly process of remanufactured parts.

No.	Assembly procedure diagram	Input						Auxiliary activities (Temperature: °C)	Value-added & control activities (Sequence of operation)	Output (Chinese)
		Assembly object	Space-Time Rules	Assembly requirements						
				Shape Size (*mm*)	Roughness (*um*)	Heat Treatment Type	Precision level			
AF1		Rolling Bearing	As shown in the figures, the rules are listed one by one based on the rules.	25 Φ40	0.8	1	7–9	Temperature [80–120], Humidity RH ≤ 70%, Shaft, MK1030, Circlip, Clamp pliers, Hammer	Clean - Lubricate - Press – Check, Low	***Rolling, Stable***
AF2		Coupling		40 Φ40	0.8	1	6–7	Temperature [16–30], Humidity RH ≤ 70%, Shaft, Flat key, Hammer	Clean - Lubricate - Press - Screw – Check, Low	***Transmission, Continuous***
AF3		Valve Core		45 Φ40	1.6	1	10–12	Temperature [0–70], Humidity RH ≤ 70%, Screwdriver, Screw, Valve ball, Hammer	Clean - Lubricate - Align - Screw – Measure, Medium	***Guided, Firm***
AF4		Gear		25 Φ45	1.6	1	10–11	Temperature [16–30], Humidity RH ≤ 70%, location pin, Shaft, Hammer	Clean - Lubricate - Press - Correct – Check, Medium	***Transfer, Axial***
AF5		Steering Engine		18 Φ32	0.4	1	5–6	Temperature [10–70], Humidity RH ≤ 70%, Shaft, Screws, Calipers, Screwdrivers	Clean - Install - Correct – Check, High	***Stable, Engaged***
AF6		Loop Bar		45 Φ36	6.3	0	11–13	Temperature [16–30], Humidity RH ≤ 70%, Support, Nut	Clean - Lubricate - Position - Rotate –Measure, Very low	***Fixed, Screw thread***
AF7		Column Sleeve		62 Φ22	3.2	0	7–8	Temperature [60–120], Humidity RH ≤ 70%, Fixed seat, File, Hammer	Clean - Lubricate - Put - Correct - Insert – Test, High	***Mobile, Smooth***
AF8		Leading Block		60 Φ15	1.6	0	9–12	Temperature [16–30], Humidity RH ≤ 70%, Set rod, Hammer	Clean - Lubricate - Counter - Pressure – Measure, Medium	***Restrict, prevent***
AF9		Cylindrical Dial		40 Φ30	1.6	1	7–11	Temperature [16–30], Humidity RH ≤ 70%, Shaft, Friction sheet, Sleeve, Sheath, Caliper	Clean - Lubricate - Align – Move, High	***Smooth, Provide***
AF10		Axis of Rotation		25 Φ44	0.8	1	9–11	Temperature [16–30], Humidity RH ≤ 70%, Shaft seat, Spring, Spring clamp	Clean - Lubricate - Center - Press - Fix – Check, Medium	***Rotate, Free***
AF11		Thrust Lever		30 Φ30	0.4	0	10–11	Temperature [16–30], Humidity RH ≤ 70%, Bearing, Spring, Spring clamp, Caliper	Clean - Lubricate - Align - Press - Debug, High	***Rotate, Transport***
AF12		Buffer Block		58 43 22	0.8	0	11–12	Temperature [100–120], Humidity RH ≤ 70%, Fixed seat, Hammer	Clean - Put in - Calibrate - Repair - Insert - Test, Medium	***Move, Interfere***

(*continued on next page*)

Table 1 (*continued*)

No.	Assembly procedure diagram	Input						Auxiliary activities (Temperature: °C)	Value-added & control activities (Sequence of operation)	Output (Chinese)
		Assembly object	Space-Time Rules	Assembly requirements						
				Shape Size (*mm*)	Roughness (*um*)	Heat Treatment Type	Precision level			
AF13		Column Sleeve		72 Φ18	1.6	1	10–12	Temperature [16–30], Humidity RH ≤ 70% Sleeve rod, Hammer, positioning pin	Clean - Lubricate - Orient - Center - Press - Measure, Low	***Merge, Sturdy***
AF14		Rolling Slider		78 28 34	1.6	0	9–12	Temperature [16–30], Humidity RH ≤ 70% Screws, slider support plate, screwdriver	Clean - Lubricate - Move - Screw - Commission - Check, Medium	***Free, Translate***

the set theory of Equation (17) as 1-log2 (1 + (3 + 1)/6 = 0.2630).

The value-added activity focuses on the calculation of the degree of similarity between the operations of the work steps in the two different procedures in a text sequence. By utilizing Equations (19)–(24), the resulting similarity value is 0.9000.

The control activity is mainly measured in the fuzzy level language, and the control level of both processes is “low” with a value of 1.0000. The keywords “rolling/滚动", “stable/稳定" and “transfer/传递" and “continuous/连续" in the output process constituent elements are substituted into Equation (26) through the GUI (Appendix C)calculation of the “HowNet” platform, and the result is 0.2416. AF1 and other procedures are calculated in a similar way. The detailed calculations are shown in Table 2.

The importance of each type of process constituent elements in the process activities varies. According to the experience of experts and the specific content described by each type of elements, the classical Analytic Hierarchy Process (AHP) can be used to determine the importance of four types of process elements, and the judgment matrix is as follows:

$$\begin{pmatrix} 1 & 4 & 3 & 5 \\ \frac{1}{4} & 1 & \frac{1}{3} & 3 \\ \frac{1}{3} & 3 & 1 & 4 \\ \frac{1}{5} & \frac{1}{3} & \frac{1}{4} & 1 \end{pmatrix}$$

The weights of I, AA, VC, and O are obtained as (0.5292,0.1342,0.2681,0.0684). Consistency ratio $CR = 0.0669 < 0.1$; the consistency test is passed.

Likewise, the judgment matrix constructed from the weights of assembly objects, space-time rules and assembly requirements in the input process constituent elements is

$$\begin{pmatrix} 1 & \frac{1}{3} & \frac{1}{2} \\ 3 & 1 & 3 \\ 2 & \frac{1}{3} & 1 \end{pmatrix}$$

Table 2
Similarity of different part assembly procedures.

Group No.	Assembly object	Space rules	Time rules	Shape Size	Roughness	Heat Treatment	Precision level	Environment	Resources	Value-added	Control	Output
(AF1, AF2)	0.6875	0.8333	0.9468	0.7500	1.000	1.0000	0.8837	0.2596	0.2630	0.9000	1.000	0.2413
(AF1, AF3)	0.5740	0.7500	0.4254	0.6666	0.5000	1.0000	0.8605	0.4583	0.0931	0.4090	0.3333	0.2179
(AF1, AF4)	0.6944	0.6667	0.6347	0.6333	0.5000	1.0000	0.9767	0.2596	0.2630	0.6522	0.3333	0.0953
(AF1, AF5)	0.3333	0.6000	0.4254	0.9167	0.5000	1.0000	0.7442	0.4545	0.0931	0.16667	0.4000	0.1506
(AF1, AF6)	0.2500	0.5000	0.4254	0.6666	0.1270	0.0000	0.7209	0.2596	0.0000	0.5000	0.0667	0.2516
(AF1, AF7)	0.2777	0.2500	0.4254	0.3833	0.2500	0.0000	0.9767	0.8333	0.1069	0.9000	0.3333	0.3988
(AF1, AF8)	0.5904	1.0000	0.9468	0.58333	0.5000	0.0000	0.8372	0.2596	0.1255	0.6667	0.0667	0.1603
(AF1, AF9)	0.5428	0.8333	0.6347	0.7500	0.5000	1.0000	0.9070	0.2596	0.0825	0.5000	0.3333	0.2914
(AF1, AF10)	0.7642	1.0000	0.6347	0.6833	0.5000	1.0000	0.9535	0.2596	0.1255	0.8333	0.0667	0.3413
(AF1, AF11)	0.7428	1.0000	0.6347	0.7500	0.5000	0.0000	0.9767	0.2596	0.2224 0.2224	0.5000	0.3333	0.1893
(AF1, AF12)	0.1851	0.1250	0.4254	0.4500	1.0000	0.0000	0.8837	0.7500	0.1255	0.1250	0.0667	0.2845
(AF1, AF13)	0.2292	1.0000	0.9468	0.2167	0.5000	1.0000	0.8605	0.2596	0.1069	0.5455	1.0000	0.3214
(AF1, AF14)	0.2952	0.1667	0.4254	0.1667	0.5000	0.0000	0.8372	0.2596	0.0000	0.5455	0.3333	0.3758

The weights of the three are (0.1571,0.5936,0.2493) CR = 0.0158 < 0.0462; the consistency test is satisfied.

The base importance levels for typical assembly requirements are set to 0.25,0.25,0.25,0.25, respectively.

The judgment matrix for spatial rules and temporal rules, as well as value-added and control activities, are represented by

$$\begin{pmatrix} 1 & 3 \\ \frac{1}{3} & 1 \end{pmatrix} \text{ and}$$

Their weights are respectively (0.7500,0.2500). CR=NaN, and they all pass the consistency test.

According to Step2-Step5, the similarity values of the multi-source information for the four types of process constituent elements and the correlation coefficients after fusion are calculated. The results are obtained in Table 3.

To evaluate the effectiveness of the proposed method, we employed three different techniques for comparison in our study, applied to 14 remanufacturing part assembly process cases. These three techniques are Word Overlap, Jaccard, and Latent Semantic Analysis (LSA). This is consistent with the approach proposed by Renu and Mocko (2016) for describing reuse assembly.

(1) Word overlap similarity is the ratio of the number of common terms in the process text to the sum of the number of words in the query text. This can be mathematically expressed

$$\textit{Word Overlap} = \frac{A \cap B}{A}$$

A is the procedure instruction to be retrieved, and B is the text of the procedure instruction to be queried in the database.

(2) The Jaccard method focuses on comparing the intersection of words in two different procedure instructions and the concatenation of words, with the ratio of the former to the latter being the similarity Jaccard score, calculated as follows

$$\textit{Jaccard similarity} = \frac{A \cap B}{A \cup B}$$

(3) Latent Semantic Analysis maps words and documents to the latent semantic space, thus removing some "noise" from the original vector space. This enhances features' robustness and improves information retrieval accuracy. It aims to extract semantic relationships between words through techniques such as singular value decomposition, representing words and texts based on this underlying semantic structure. The process can be summarized as

Step 1: Analyze the different process contents in the assembly procedure instructions and construct a vocabulary-text matrix ζ.

Step 2: Perform singular value decomposition on the lexical-textual matrix.

Step3: Dimensionality reduction of the matrix using the singular value decomposition method.

Step 4: Construct the latent semantic space using the reduced dimensional matrix.

Fig. 10 shows the comparison of the similarity degree between the 11 groups of part assembly procedures calculated by this method and the other three methods.

Fig. 11 illustrates the results of a double-Y comparison, displaying the mean and variance of the similarity calculated by the four distinct methods corresponding to Fig. 10. The findings indicate that the method introduced in this paper demonstrates a significant level of similarity among different procedures, with a variance value of 0.06. In comparison to other methods that solely examine the superficial attributes of process instructions, the approach presented in this paper exhibits certain advantages.

(i) This paper systematically investigates a method based on process constituent elements, providing a convincing analysis. It explores the nuanced requirements and local features embodied within the process, utilizing multi-source information derived from remanufacturing process instructions. The study illuminates the commonalities among diverse process instructions through an examination of embedded procedure features. This comprehensive research approach yields persuasive results that deepen our understanding of similarities in process instructions.

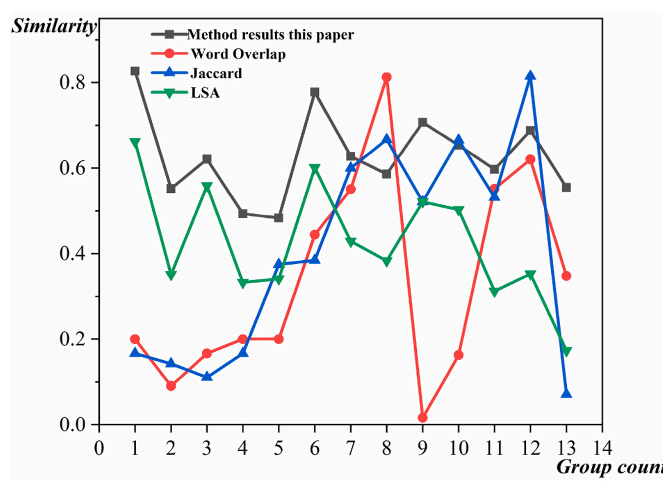


Fig. 10. Similarity values of different methods.

Table 3
Similarity and correlation coefficients of process constituent elements of AF1 and other parts of the assembly procedures.

No.	Input	Auxiliary Activities	Value-added Control Activities	Output	Relation Coefficient
(AF1,AF2)	0.4438349	0.0351805	0.2479925	0.0165049	0.8267
(AF1,AF3)	0.3394435	0.0247465	0.1045791	0.0149044	0.5523
(AF1,AF4)	0.3508988	0.0351805	0.1534805	0.0065185	0.6206
(AF1,AF5)	0.2723124	0.024619	0.0603232	0.010301	0.4936
(AF1,AF6)	0.2339737	0.0087096	0.1050081	0.0172094	0.4833
(AF1,AF7)	0.1577433	0.0387167	0.2033069	0.0272779	0.7777
(AF1,AF8)	0.4881283	0.0213412	0.1385273	0.0109645	0.6275
(AF1,AF9)	0.3906298	0.0170132	0.1228769	0.0199318	0.5859
(AF1,AF10)	0.4614024	0.0213412	0.1720264	0.0233449	0.7069
(AF1,AF11)	0.4596232	0.0310941	0.1228769	0.0129481	0.6531
(AF1,AF12)	0.1107782	0.0377941	0.0296049	0.0194598	0.5970
(AF1,AF13)	0.4580991	0.0194691	0.1767114	0.0219838	0.6877
(AF1,AF14)	0.133031	0.0087096	0.1320258	0.0257047	0.5548

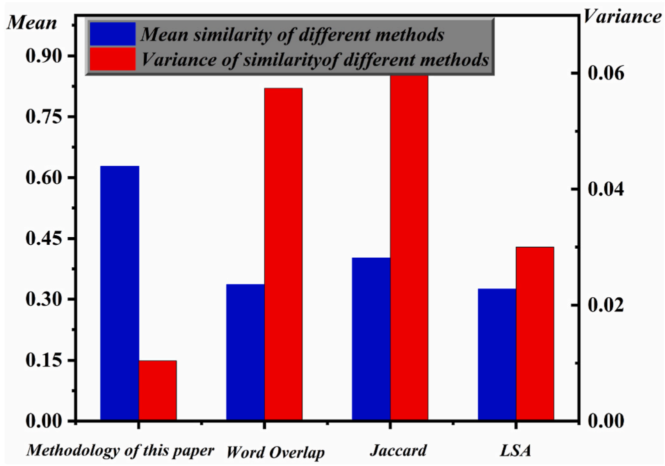


Fig. 11. Histogram of mean and variance statistics of similarity of different methods.

(ii) This paper critically evaluates a method based on processing text instructions, primarily employing word overlap, Jaccard, and latent semantic analysis to discern similarities among procedures. The effectiveness of these methods depends heavily on the design quality of the remanufacturing process instructions, neglecting the implicit information within them. As illustrated in Figs. 10 and 11, relying solely on surface information in the process instructions to tackle reuse issues (partial features) can result in significant fluctuations in outcomes. These fluctuations are evidenced by variations in mean and variance values derived from individual similarities.

(iii) This evaluates the design and composition of assembly instructions for remanufactured parts, taking into consideration the influence of human subjectivity and habitual expressions. The research methodology proposed in the referenced literature (Renu and Mocko, 2016) emphasizes the need for detailed specifications and precise design in process instructions. However, it seemingly overlooks the importance of word order, which could potentially affect the precision of semantic expression. This observation highlights the necessity for further research to enhance the semantic accuracy in the design of process instructions within the remanufacturing domain.

7. Conclusion and prospect

The study asserts the critical role of the remanufacturing assembly process in determining the quality of remanufactured products. A novel approach based on the process constituent elements model is introduced, providing a comprehensive analysis of the assembly process by incorporating key feature information as a gene sequence. This contributes to the timely identification and retrieval of key fine-grained information in assembly process information. The model incorporates a reusability and retrieval system for different assembly procedures, facilitating the accelerated design process of new assembly techniques while conserving resources. The study introduces a novel model for process analysis and optimization, providing a more advanced and systematic approach to research on remanufacturing processes. It enriches the understanding of fine-grained remanufacturing process retrieval and reuse. Overall, this study play an important role in advancing research in the field of remanufacturing and provides valuable insights and directions for future studies.

This paper emphasizes the importance of establishing robust and collaborative methods for designing, sharing and retrieving the assembly process for remanufacturing. However, there may be initial performance differences caused by the remanufacturing production process when comparing remanufactured parts, directly useable parts, and new parts in the remanufacturing assembly. Neglecting this difference can potentially affect the reuse of fine-grained information in the remanufacturing assembly process. Furthermore, although the allocation of weights to the process constituent elements in the system is typically based on experience and knowledge, complex remanufacturing problems and limited cognitive abilities can lead to bias. Further research and empirical validation are needed to overcome the limitations of the current study to enhance the reliability of the conclusions.

CRediT authorship contribution statement

Zhongyi Wu: Writing – original draft, Methodology, Conceptualization, Formal analysis. **Weidong Liu:** Funding acquisition, Conceptualization, Resources, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data availability

Data will be made available on request.

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Appendix A

Notations

n	the procedure is composed of n work steps	m	each work step can be divided into m specific characteristics of the process constituent elements
$\oplus$	the parallel base at the same level	$\otimes$	the arrangement of base pairs
AA_{obj}	assembly object base pair	AA_{req}	assembly requirement base pair
SP_{lev}	spatial hierarchy rule base pair	SP_{pos}	position relation base pair
Ti_{seq}	the assembly sequence base pair	Ti_{adj}	assembled adjacency relation base pair
Ti_{con}	spatial constraint relation base pair	WS_{Vh}	Value-added activity attributes of work steps
WS_{cv}	Control activity properties of the work steps	$WS_{vh_{attr}}$	the base pair sequence of the value-added activity
θ_{hr}	the weight of the r-th work step activity	WS_{ch_attr}	The control activity base pair sequence

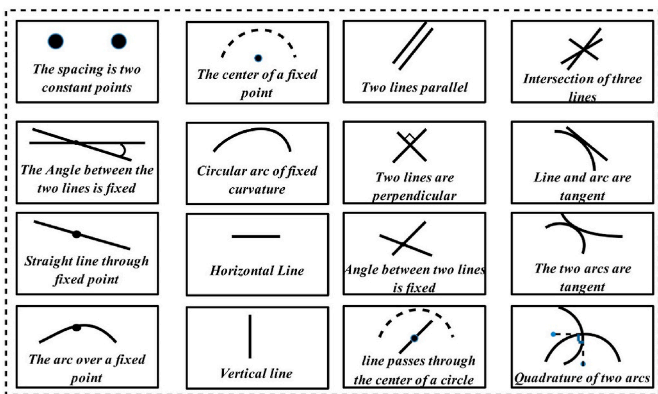
(continued on next page)

(*continued*)

n	the procedure is composed of n work steps	m	each work step can be divided into m specific characteristics of the process constituent elements
δ^h_{cv1}	the importance of prevention factor	$WS_{vhp}\text{-}trpeRe_{sw}$	The procedure s has w resource factors
δ^h_{cv2}	the importance of control factor	$WS_{cvi}\text{-}trpeEn_{sg}$	the s procedure has g environmental factors
ER_{si}	the expected evaluation result of the *i-th* work step activity	θ_{ERsi}	the degree of influence of the i-th work step activity
Fun_{Num}	common functional features of the two assemblies	A^{lFun}_{obj}	
L_i, L_j	the length values of the two parts	$max(L)$	The maximum length values of the batch parts
$min(L)$	The minimum length values of the batch parts	Ra	the roughness value of a part in a new procedure
Ra_H	The highest roughness values of the parts to be compared	Ra_L	The lowest roughness values of the parts to be compared
$u_{Ra}(x_i)$	the fuzzy membership function of the roughness of a part in a procedure concerning respect to the new procedure	h	the minimum depth of the LCS of the hierarchical structure
$\delta(i,j)$	The score function	$Sim(x_1, x_2)$	the similarity match of the value-added activities of the two procedures
$Sim(e_{ij}, e_{0j})$	the similarity between manufacturing procedure i and procedure 0	$C(s_j)$	The similarity between the *j-th* procedure to be retrieved and the target procedure.

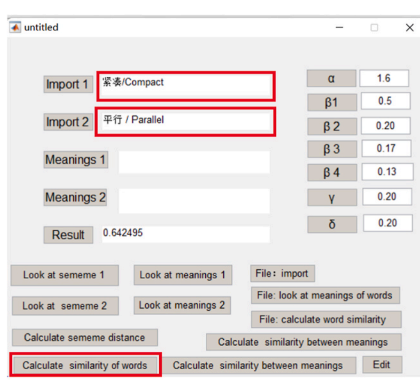
Appendix B. Assembly geometry constraints in the input process constituent elements

In the typical 2-dimensional contour of structural parts, the following figure provides a summary of the geometric constraint types for these (Zhao et al., 2013).



Appendix C. GUI platform for phrase similarity calculation

The similarity between phrases h_{1i} and h_{2i} is mainly calculated through the platform of Chinese “HowNet”, which comprehensively considers the similarity between phrases from three aspects of sense origin, set and feature structure, and on this basis. The following figure shows the GUI interface developed based on the principle, just enter h_{1i} and h_{2i} in the red box and click “Calculate similarity of words” to get the similarity value.



Note: The Arabic letters α $\beta_1, \beta_2, \beta_3, \beta_4, \gamma$ and δ in the GUI interface indicate the different semantic weights in ‘HowNet’ (Huang, et al.2014) .

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